

Time Series Analysis of Brent Crude Oil Prices

Modeling and Forecasting Monthly Prices Using Box-Jenkins ARIMA Models
(June 1987 – February 2026)

This study applies the Box-Jenkins methodology to 465 monthly observations of Brent Crude Oil spot prices sourced from FRED. Following variance stabilization via Box-Cox transformation and stationarity testing, an ARIMA(0,1,1) model is selected and evaluated on a 46-month out-of-sample test set, with implications discussed for oil-dependent economies such as Algeria.

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Academic Year: 2025–2026

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Abstract

This project applies the Box-Jenkins methodology to model and forecast monthly Brent Crude Oil spot prices from June 1987 to February 2026 (465 observations) using the Federal Reserve Economic Data (FRED) platform. To address the characteristics of commodity price series, a Box-Cox transformation with an optimal parameter of $\lambda = -0.3$ is utilized to stabilize the variance, followed by first-order differencing to achieve stationarity. Five candidate models — ARIMA(0,1,1), ARIMA(1,1,1), ARIMA(2,1,1), ARIMA(1,1,0), and ARIMA(2,1,0) — were identified through Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) analysis. Model selection was conducted using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), leading to the selection of ARIMA(0,1,1) based on the Principle of Parsimony. Model adequacy was verified using rigorous residual diagnostics, including the Ljung-Box test for autocorrelation, the Runs test for independence, and the Shapiro-Wilk test for normality. While independence and uncorrelation conditions are satisfied, normality is rejected due to heavy tails generated by extreme global shocks. Out-of-sample forecast performance was evaluated on a withheld 10% test set spanning May 2022 to February 2026. The results demonstrate that while the model captures uncertainty, it produces a flat forecast that systematically overestimates prices during the post-pandemic market correction, highlighting the limitations of pure linear ARIMA models under structural breaks. The policy implications of these limitations are discussed in the context of oil-dependent economies like Algeria.

1 Introduction

Brent Crude Oil is a major global benchmark, representing light, sweet crude oil extracted from the North Sea. It is used to price approximately 70% of the world's internationally traded crude oil. Because crude oil is a primary energy source and chemical feedstock, movements in Brent prices directly influence fuel costs, transportation, food prices, and manufacturing costs worldwide.

For oil-exporting countries, oil price fluctuations have direct macroeconomic consequences. In Algeria, oil and gas exports represent the overwhelming majority of government revenues and export earnings. Consequently, sudden changes in the Brent price exert significant pressure on the national budget, exchange rates, and public investment plans. Accurately modeling and forecasting this series is of key interest for national economic planning and budget risk management.

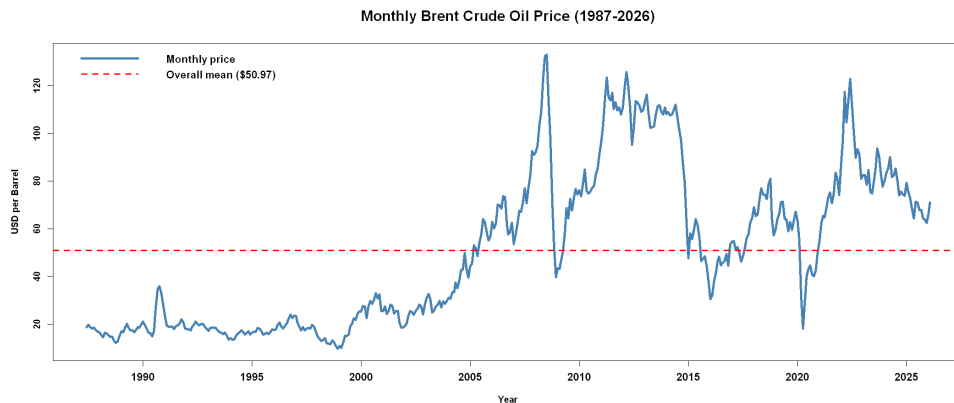


Figure 1: Monthly Brent Crude Oil Price (June 1987 – February 2026)

The dataset analyzed in this study consists of 465 monthly observations of the Brent Crude Oil spot price in US dollars (USD) per barrel, spanning June 1987 to February 2026. The source of the data is the Federal Reserve Economic Data (FRED) database. Over this nearly 40-year period, the oil market was subjected to several extreme global events:

- The 1990–1991 Gulf War oil price spike.
- The 1997–1998 Asian financial crisis, during which prices dropped to a historical minimum of \$9.82 per barrel.
- The 2008 global commodity boom and subsequent subprime financial crisis, where prices peaked at a historical maximum of \$132.72 per barrel, followed by a rapid collapse.
- The 2020 COVID-19 pandemic demand shock, which saw oil prices collapse to historic lows.
- The 2022 Russia-Ukraine war, which caused a sharp spike in energy prices, followed by a prolonged downward market correction.

This research systematically develops an Autoregressive Integrated Moving Average (ARIMA) model to capture the underlying stochastic process of the series. We present data preparation, stationarity and transformation pipelines, model selection, diagnostic validation, and out-of-sample forecasting performance.

2 Model Specification

This section describes the identification and specification of candidate ARIMA models. The process includes data preparation, visual inspection of the raw series, variance stabilization, stationarity testing, and lag order identification via ACF/PACF and subset regression.

2.1 Data Preparation and Time Series Object Creation

The raw dataset was loaded and checked for integrity. The temporal index is complete with no missing observations. The sequence contains exactly 465 monthly observations from June 1987 to February 2026. A summary of the dataset is presented below:

- **Number of observations:** 465 months.
- **Minimum price:** \$9.82 per barrel (December 1998).
- **Maximum price:** \$132.72 per barrel (July 2008).
- **Mean price:** \$50.97 per barrel.

The series was converted into a formal R time series object ('ts' class) with a monthly frequency (*frequency* = 12) starting in June 1987 (*start* = *c*(1987, 6)).

2.2 Visual Inspection

Visual inspection of the raw series (Figure 1) reveals two distinct behaviors:

1. **Non-constant mean:** The series displays prominent trends with no fixed baseline, showing that the mean changes over time.
2. **Non-constant variance (Heteroscedasticity):** The amplitude of price fluctuations is much larger in the latter half of the sample (post-2000, where prices fluctuated between \$30 and \$130) compared to the first half (1987–2000, where prices remained stable between \$10 and \$40). The magnitude of the volatility appears to grow with the overall price level, indicating multiplicative variance.

2.3 Train / Test Split

To perform a rigorous out-of-sample evaluation, the dataset is divided into a training and a testing set using a 90/10 split:

- **Training Set:** June 1987 to April 2022 (419 observations).
- **Testing Set:** May 2022 to February 2026 (46 observations).

This partition is chosen instead of a standard 95/5 split to provide a larger out-of-sample window (46 months). Crucially, the test period covers the high-volatility phase following the outbreak of the Russia-Ukraine war in early 2022 and the subsequent correction, creating a realistic forecasting challenge. All model identification and estimation steps are conducted strictly on the training set.

2.4 Variance Stabilization (Box-Cox Transformation)

To address the heteroscedasticity observed in the visual inspection, the Box-Cox family of power transformations is evaluated:

$$T(Y_t) = \begin{cases} \frac{Y_t^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \ln(Y_t) & \lambda = 0 \end{cases} \quad (1)$$

The optimal parameter λ is determined by maximizing the log-likelihood function over the training series. The profile search yields:

$$\text{Optimal } \lambda = -0.3$$

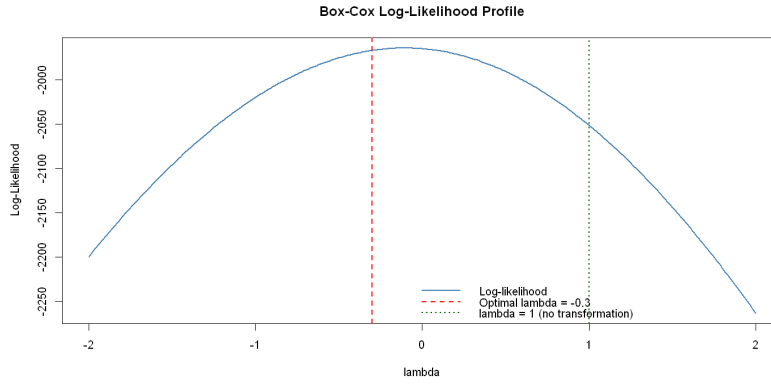


Figure 2: Box-Cox Log-Likelihood Profile

As shown in Figure 2, the log-likelihood peak lies at $\lambda = -0.3$. This value is statistically different from $\lambda = 1$ (which represents no transformation), justifying the need for a transformation. Because $\lambda = -0.3$ is close to 0, it behaves similarly to a log transformation, which is standard for financial asset price series.

Applying this transformation stabilizes the variance across the series, as illustrated in the comparison in Figure 3. The amplitude of fluctuations in the transformed series is uniform across the entire range.



Figure 3: Original vs Box-Cox Transformed Training Series ($\lambda = -0.3$)

2.5 Stationarity and Differencing

Although the Box-Cox transformation successfully stabilized the variance, the transformed series (Z_t) still contains a strong trend and is non-stationary. To confirm this formally, the Augmented Dickey-Fuller (ADF) unit root test is applied to Z_t . The test yields:

$$\text{ADF statistic} = -2.4529, \quad \text{Lag Order} = 7, \quad p\text{-value} = 0.3861$$

Since the p -value (0.3861) is well above the significance level of $\alpha = 0.05$, we fail to reject the null hypothesis of a unit root. This confirms that the transformed series is non-stationary.

To achieve stationarity, first-order differencing ($d = 1$) is applied:

$$W_t = \Delta Z_t = Z_t - Z_{t-1} \quad (2)$$

The differenced transformed series (W_t) is plotted in Figure 4. The series fluctuates around a constant mean of zero, with no trend.

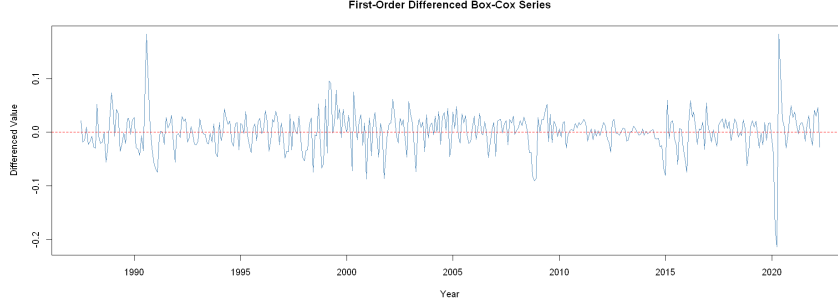


Figure 4: First-Order Differenced Box-Cox Transformed Series

An ADF test is run on the differenced series W_t to confirm stationarity:

$$\text{ADF statistic} = -8.0772, \quad \text{Lag Order} = 7, \quad p\text{-value} \leq 0.01$$

The p -value is less than 0.01 (rejecting the null hypothesis of a unit root), which confirms that the series is stationary. Thus, the integration order for our model is set to $d = 1$.

2.6 ACF and PACF Analysis

To identify the appropriate Autoregressive (p) and Moving Average (q) orders, the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) are computed on the stationary series W_t (Figure 5).

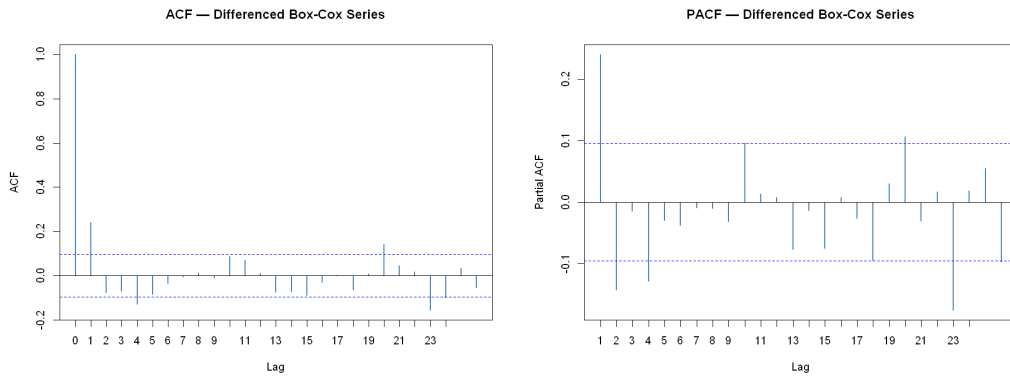


Figure 5: ACF and PACF of the Differenced Box-Cox Series

The plots reveal the following properties:

- **ACF:** A significant negative spike is present at lag 1, after which all values fall within the 95%

confidence bands (with the exception of isolated, irregular crossings at higher lags like lag 20 and 23). This cut-off behavior at lag 1 is typical of a Moving Average process of order 1, or MA(1).

- **PACF:** Significant negative spikes occur at lags 1 and 2, followed by a gradual decay towards zero. This tailing-off behavior indicates an Autoregressive process.

Based on these observations, candidate models are identified within the ARIMA($p, 1, q$) framework. In addition to the simple models, mixed models are included to account for possible interactions:

1. **ARIMA(0,1,1):** Pure MA(1) process on the differenced series. Highly parsimonious, motivated by the ACF cutting off at lag 1.
2. **ARIMA(1,1,0):** Pure AR(1) process as a standard benchmark.
3. **ARIMA(2,1,0):** Pure AR(2) process, motivated by the PACF spikes at lags 1 and 2.
4. **ARIMA(1,1,1):** Mixed ARMA(1,1) process, allowing both AR and MA effects.
5. **ARIMA(2,1,1):** Mixed ARMA(2,1) process, capturing the two PACF spikes and the ACF spike.

To supplement this analysis, subset regression was performed using the ‘armasubsets’ function up to lag 12 (to account for annual seasonality in monthly data). However, the subset search returned a warning of linear dependencies among the lagged regressors, indicating that the subset search is unstable for this volatile commodity series. Consequently, the selection of candidate models relies on the ACF/PACF analysis.

2.7 Robustness Check on Restricted Time Window (2009–2018)

To verify that the identified lag structures are stable, a robustness check was performed by repeating the identification pipeline on a restricted sub-period covering **January 2009 to December 2018** (120 observations). This window is chosen because it represents a stable market regime, excluding the two massive external shocks: the 2008 subprime crisis collapse and the 2020 COVID-19 pandemic shock.

The restricted series is transformed ($\lambda = -0.3$), differenced ($d = 1$), and confirmed stationary via the ADF test (statistic = -4.5118, p -value = 0.01). The ACF and PACF are plotted in Figure 6.

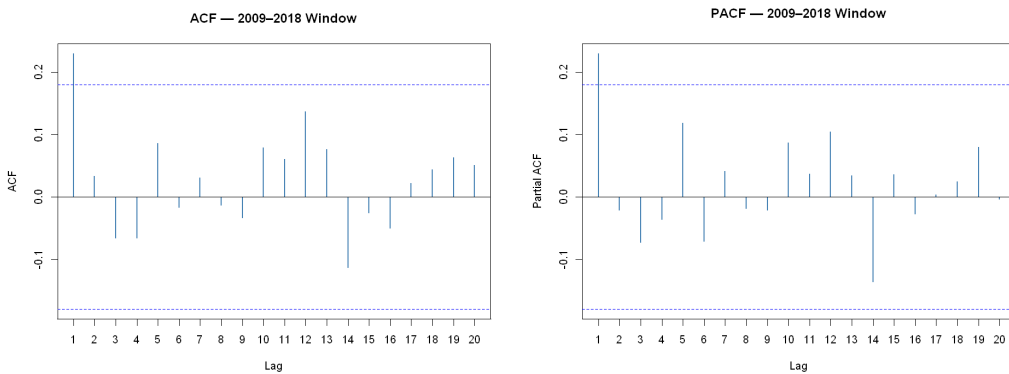


Figure 6: ACF and PACF of the Restricted Window Series (2009–2018)

The plots show:

- **ACF**: A single significant negative spike at lag 1, cutting off immediately.
- **PACF**: A single significant spike at lag 1, decaying thereafter.

This restricted analysis confirms that the underlying process is governed by a simple low-order structure (primarily lag 1). The high-lag crossings observed in the full training series (e.g. lags 20 and 23) do not reappear in this stable period, confirming that they are artifacts of the extreme outliers of 2008 and 2020 rather than true systematic dependencies. This result supports our choice of low-order candidate models.

3 Model Fitting and Diagnostics

This section presents the parameter estimation for the five candidate models, their comparison using information criteria, and diagnostic tests on their residuals.

3.1 Model Selection and Estimation

Each candidate model was fitted on the Box-Cox transformed training series using Maximum Likelihood Estimation (MLE). The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are summarized in Table 1.

Table 1: Information Criteria Comparison of Candidate Models

Model	AIC	BIC	Parameters
ARIMA(0,1,1)	-1632.49	-1624.42	1 (MA)
ARIMA(1,1,0)	-1625.79	-1617.71	1 (AR)
ARIMA(2,1,0)	-1632.46	-1620.36	2 (AR)
ARIMA(1,1,1)	-1632.58	-1620.48	2 (AR, MA)
ARIMA(2,1,1)	-1633.72	-1617.58	3 (AR, MA)

The information criteria select different models:

- **AIC** selects the richer **ARIMA(2,1,1)** model (AIC = -1633.72), as it imposes a smaller penalty for model complexity.
- **BIC** selects the simpler, more parsimonious **ARIMA(0,1,1)** model (BIC = -1624.42), as its penalty for additional parameters is more stringent.

Following this result, both models are estimated and analyzed to select the final model.

3.2 Parameter Estimates

The MLE parameter estimates, standard errors, and corresponding t -statistics for the two candidate models are presented in Table 2.

Table 2: Maximum Likelihood Parameter Estimates

Model	Parameter	Coefficient	Std. Error	t-statistic	p-value
ARIMA(0,1,1)	$ma_1 (\theta_1)$	0.3026	0.0505	5.987	< 0.001
ARIMA(2,1,1)	$ar_1 (\phi_1)$	0.9252	0.1655	5.591	< 0.001
	$ar_2 (\phi_2)$	-0.2836	0.0494	-5.740	< 0.001
	$ma_1 (\theta_1)$	-0.6721	0.1696	-3.964	< 0.001

All estimated coefficients in both models are statistically significant, with absolute t -statistics exceeding the critical threshold of 1.96 (corresponding to p -values < 0.001).

The mathematical formulations of the estimated models are defined as follows:

- **ARIMA(0,1,1):**

$$(1 - B)Z_t = (1 + 0.3026B)\varepsilon_t \implies W_t = \varepsilon_t + 0.3026\varepsilon_{t-1} \quad (3)$$

- **ARIMA(2,1,1):**

$$(1 - 0.9252B + 0.2836B^2)(1 - B)Z_t = (1 - 0.6721B)\varepsilon_t \quad (4)$$

$$\implies W_t = 0.9252W_{t-1} - 0.2836W_{t-2} + \varepsilon_t - 0.6721\varepsilon_{t-1} \quad (5)$$

where $Z_t = \frac{Y_t^{-0.3} - 1}{-0.3}$ represents the transformed series, B is the backshift operator ($BX_t = X_{t-1}$), and $\varepsilon_t \sim WN(0, \sigma^2)$ is a white noise process.

3.3 Residual Diagnostics

To evaluate model adequacy, the residuals of both fitted models are subjected to rigorous diagnostic checks to verify whether they satisfy the classical white noise assumptions: homoscedasticity, independence, and normality.

3.3.1 Standardized Residuals and Homoscedasticity

The standardized residuals for both models are plotted in Figure 7.

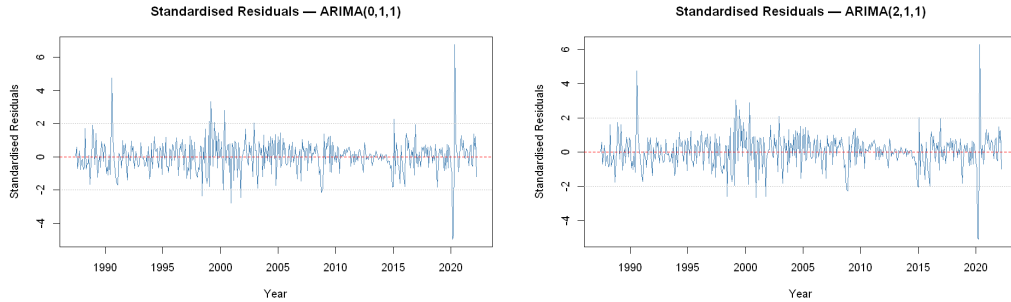


Figure 7: Standardized Residuals for ARIMA(0,1,1) and ARIMA(2,1,1)

Both series fluctuate around a mean of zero with no systematic trend or visible changes in variance. The majority of observations fall within the horizontal bands of ± 2 . However, two large outliers are present in both plots:

1. A large positive shock in 1991 (corresponding to the Gulf War price spike).
2. A large negative shock in 2020 (corresponding to the COVID-19 demand crash).

These outliers represent structural shifts from external shocks that are not captured by standard linear models.

3.3.2 Residual Autocorrelation (ACF and Ljung-Box Test)

The residual ACF plots are shown in Figure 8.

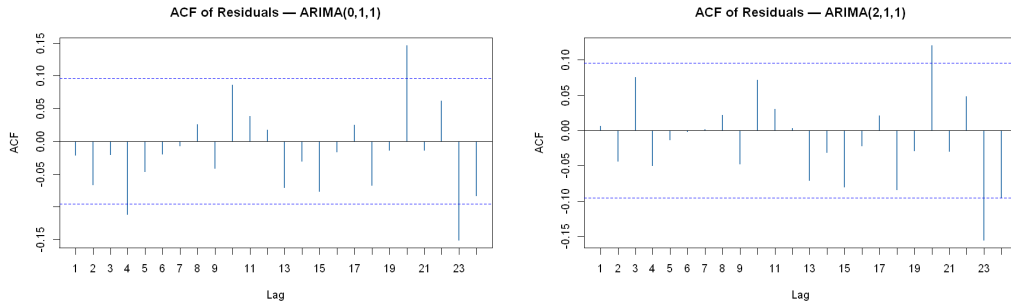


Figure 8: ACF of Residuals for ARIMA(0,1,1) and ARIMA(2,1,1)

For both models, almost all autocorrelations fall within the significance limits. For ARIMA(0,1,1), minor crossings are observed at lags 4, 20, and 23. For ARIMA(2,1,1), only borderline crossings are visible at lags 20 and 23. This shows that both models capture the bulk of the temporal dependence in the series.

To test for remaining autocorrelation formally, the Ljung-Box Q test is conducted at lag $K = 20$. The null hypothesis H_0 states that the residuals are uncorrelated up to lag 20:

$$\text{ARIMA}(0,1,1): \chi^2 = 30.5534, \quad p\text{-value} = 0.0614$$

$$\text{ARIMA}(2,1,1): \chi^2 = 23.7938, \quad p\text{-value} = 0.2515$$

Both models yield p -values greater than the $\alpha = 0.05$ threshold, meaning we fail to reject the null hypothesis of uncorrelation. The residuals can be considered uncorrelated, confirming model adequacy in terms of temporal structure. Note that ARIMA(2,1,1) provides a larger margin ($p = 0.2515$ vs $p = 0.0614$), indicating a cleaner residual structure.

3.3.3 Residual Independence (Runs Test)

The Runs test is applied to assess whether the residuals exhibit a systematic pattern of signs (runs of positive or negative values), which would indicate remaining dependence. The null hypothesis H_0

states that the signs are independent:

$$\text{ARIMA}(0,1,1): \quad p\text{-value} = 0.4110$$

$$\text{ARIMA}(2,1,1): \quad p\text{-value} = 0.5999$$

Both p -values are well above 0.05. We fail to reject H_0 , confirming that the residuals are independent.

3.3.4 Normality Checks (Shapiro-Wilk Test and Graphical Analysis)

To check the normality assumption, histograms and Normal Q-Q plots of the residuals are generated (Figure 9).

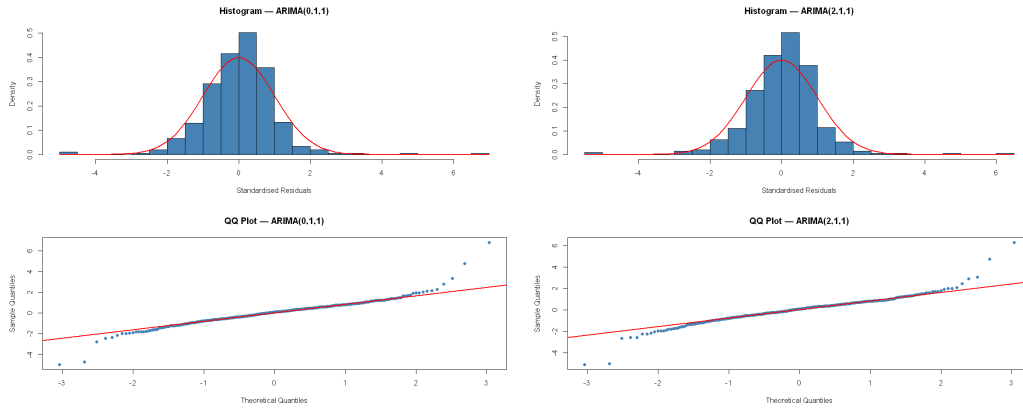


Figure 9: Histograms and Normal Q-Q Plots of Residuals

The histograms are bell-shaped and centered around zero, indicating approximate normality in the center of the distribution. However, the Q-Q plots show departures from the diagonal line at both tails. This indicates the presence of heavy tails, which are driven by the extreme positive and negative outliers identified in the residuals plots.

To test normality formally, the Shapiro-Wilk test is applied. The null hypothesis H_0 states that the residuals are normally distributed:

$$\text{ARIMA}(0,1,1): \quad W = 0.9195, \quad p\text{-value} \approx 0$$

$$\text{ARIMA}(2,1,1): \quad W = 0.9226, \quad p\text{-value} \approx 0$$

Both models reject the null hypothesis of normality ($p\text{-value} < 0.001$).

3.4 Summary of Diagnostics and Model Selection

Table 3 summarizes the diagnostic results for both candidate models.

Table 3: Diagnostic Test Summary

Diagnostic Test	ARIMA(0,1,1)	ARIMA(2,1,1)	Target
Ljung-Box Q Test (p -val)	0.0614 (Pass)	0.2515 (Pass)	> 0.05
Runs Test (p -val)	0.4110 (Pass)	0.5999 (Pass)	> 0.05
Shapiro-Wilk Test (p -val)	< 0.001 (Fail)	< 0.001 (Fail)	> 0.05
AIC	-1632.49	-1633.72	Lower is better
BIC	-1624.42	-1617.58	Lower is better

Both models satisfy the conditions for residual independence and uncorrelation. However, both models fail the normality test.

Because the normality assumption is violated, **neither model is strictly adequate under the classical Box-Jenkins ARIMA assumptions**. This limitation is common when fitting linear time series models to long economic series, where structural breaks and crisis-driven volatility generate heavy-tailed outliers.

Since both models are non-normal, the diagnostic tests cannot distinguish between them. The choice is based on the information criteria and complexity:

- ARIMA(2,1,1) contains two additional AR parameters, but yields only a minor improvement in the AIC score ($\Delta AIC = -1.23$) compared to the simpler model.
- ARIMA(0,1,1) is favored by the BIC and the restricted window analysis, which confirmed that the underlying temporal structure is simple.

Applying the Principle of Parsimony, **ARIMA(0,1,1)** is selected as the final model. We generate forecasts as a demonstration of the methodology, and interpret the results in light of these model limitations.

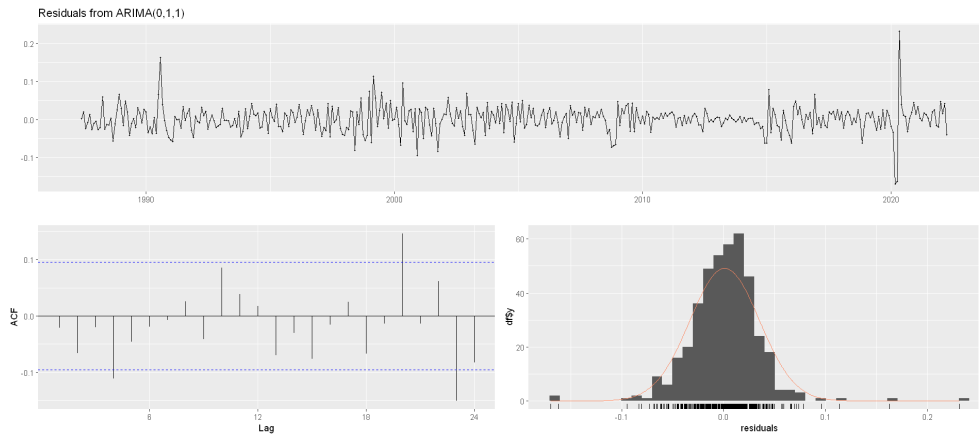


Figure 10: Final ARIMA(0,1,1) Residual Diagnostic Dashboard

4 Forecasting

This section presents the forecasting results for the test period (May 2022 to February 2026, 46 steps ahead) using the selected ARIMA(0,1,1) model.

4.1 Methodology and Back-Transformation

Using the estimated ARIMA(0,1,1) model, forecasts are generated for $h = 46$ months ahead. Since the model was fitted on the Box-Cox transformed series, the raw forecasts ($\hat{Z}_{t+h|t}$) and their standard errors ($se_{t+h|t}$) are on the transformed scale.

To obtain predictions on the original USD scale, the forecasts and their 95% prediction intervals must be back-transformed. The inverse Box-Cox transformation is defined as:

$$Y_{t+h} = (\lambda \hat{Z}_{t+h|t} + 1)^{1/\lambda} \implies Y_{t+h} = (-0.3 \hat{Z}_{t+h|t} + 1)^{-1/0.3} \quad (6)$$

The 95% prediction interval boundaries are computed on the transformed scale and then back-transformed:

$$\text{Lower 95\%} = \text{InvBoxCox} (\hat{Z}_{t+h|t} - 1.96 \cdot se_{t+h|t}) \quad (7)$$

$$\text{Upper 95\%} = \text{InvBoxCox} (\hat{Z}_{t+h|t} + 1.96 \cdot se_{t+h|t}) \quad (8)$$

4.2 Forecast Evaluation and Discussion

The actual spot prices, back-transformed predictions, and 95% prediction intervals for the test period are shown in Figure 11.

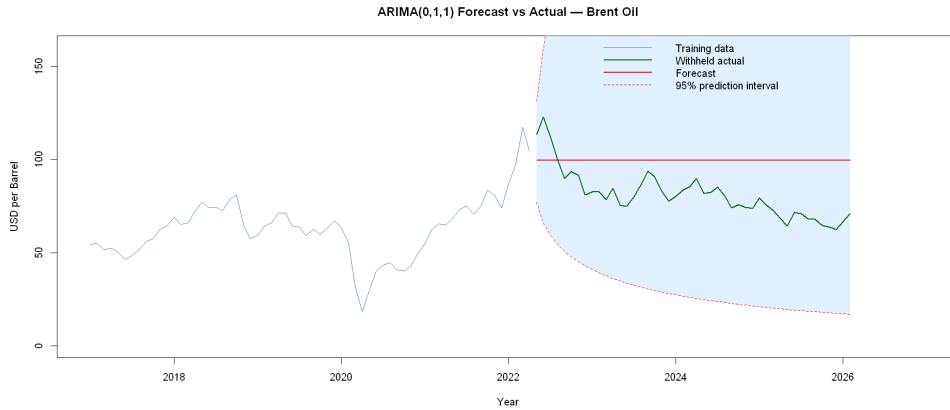


Figure 11: ARIMA(0,1,1) Forecast vs. Actual Spot Prices (May 2022 – February 2026)

The forecast behavior displays the following properties:

1. **Flat Mean Forecast:** The mean forecast is flat at a constant value of \$99.57 per barrel. Because the model contains no AR terms ($p = 0$) and a single difference ($d = 1$), the forecast beyond step 1 ($h > 1$) is constant:

$$\hat{Z}_{t+h|t} = \hat{Z}_{t+1|t} = Z_t + \theta_1 \hat{\epsilon}_t$$

The back-transformed forecast is therefore constant at \$99.57 per barrel for the entire 46-month horizon.

2. **Growing Prediction Intervals:** Due to the integrated component ($d = 1$), the forecast uncertainty grows with the horizon. The prediction intervals widen rapidly:

- **Month 1 (May 2022):** \$77.07 to \$131.42 (width = \$54.35).
- **Month 46 (Feb 2026):** \$16.90 to \$5,663.21 (width = \$5,646.31).

The upper boundary of \$5,663.21 is skewed because the inverse Box-Cox transformation is non-linear, which inflates the upper limit.

The actual oil price declined from its war-driven peak of \$113.34 in May 2022 to \$70.89 in February 2026, representing a market correction. Because the model predicted a flat path, it systematically overestimated the actual prices during the test period. The quantitative forecast accuracy metrics are summarized in Table 4.

Table 4: Out-of-Sample Forecast Accuracy Metrics

Metric	Value
Mean Error (ME)	-18.5630
Root Mean Squared Error (RMSE)	22.5004
Mean Absolute Error (MAE)	20.7426
Mean Percentage Error (MPE)	-25.6268%
Mean Absolute Percentage Error (MAPE)	27.4925%
95% Prediction Interval Coverage	46 / 46 (100%)

The metrics indicate a moderate forecast error:

- The negative Mean Error (ME = -18.56) and MPE (-25.63%) confirm that the model systematically over-forecasts.
- The Mean Absolute Error (MAE = 20.74) indicates that, on average, the forecast deviates from the actual price by \$20.74 per barrel.
- The Mean Absolute Percentage Error (MAPE = 27.49%) reflects the difficulty of predicting commodity prices during structural corrections.
- Despite these errors, all 46 observations fall within the 95% prediction interval. The model captures the range of future price outcomes, but the wide intervals limit its practical utility for forecasting.

5 Conclusion and Discussion

This project applied the Box-Jenkins ARIMA methodology to monthly Brent Crude Oil prices from 1987 to 2026. The variance was stabilized using a Box-Cox transformation with $\lambda = -0.3$, and stationarity was achieved through first-order differencing ($d = 1$). Model selection led to the parsimonious ARIMA(0,1,1) model.

Residual diagnostics showed that while the model captures the temporal structure (satisfying independence and uncorrelation conditions), it fails the normality assumption due to outliers from global shocks. These outliers are associated with major events like the 1991 Gulf War, the 2008

subprime crisis, and the 2020 COVID-19 pandemic. Under out-of-sample testing, the model's forecasts remained flat at \$99.57 per barrel, systematically overestimating the price correction that occurred after 2022.

These findings highlight the limitations of linear ARIMA models for commodity markets characterized by high volatility, geopolitical shocks, and structural breaks. Because ARIMA models assume linear dependencies and constant parameters, they cannot adapt to structural shifts, and their long-term forecasts revert to a constant value.

For oil-dependent nations like Algeria, these modeling limitations have practical consequences. Since national budget planning depends on future oil revenues, relying on flat ARIMA forecasts can lead to systematic revenue overestimation during market downturns. Underestimating price uncertainty can result in budget deficits and economic instability.

To improve forecasting performance, future research should explore extensions that address these limitations:

1. **GARCH Models:** Autoregressive Conditional Heteroscedasticity (GARCH) models can capture the volatility clustering observed in the residuals.
2. **Markov Regime-Switching Models:** These models allow the parameters to shift between high-volatility and low-volatility regimes, capturing structural breaks.
3. **Machine Learning Extensions:** Hybrid models combining ARIMA with Long Short-Term Memory (LSTM) networks or neural networks may help capture non-linear dynamics and improve forecasting accuracy during market corrections.